

FT85KS Manual V1.5

Specifications :

- Firmware: V1.5 or above (firmware update via Flipsky ESC Tool supported)
- Voltage: 14V-85V (safe for 4S-20S)
- Continuous current: 250A (Values depend on the mounting, ambient temperature and air/water circulation around the device!);
- Peak Current: 400A
- BEC: 5V@1A
- Programmable Tool: Flipsky ESC TOOL
- PWM Frequency Control: YES
- Multiple Speed Gear Limit Control: YES (in UART / ADC mode)
- Regenerative Capacity: YES
- Communication Ports: USB, CAN, UART, PPM, ADC
- Input Signals Support: PPM, ADC, PAS, UART (compatible with Flipsky remote such as VX1, VX1Pro, VX2, VX3, VX3Pro, VX4Pro)
- Startup Modes Support: IPDS (sensorless), HFPI (sensorless), HALL (sensored), ENCODER (ABI Encoder, AS5047 Absolute Type Encoder SPI Port), OPEN (Sensorless Open-Loop Startup)
- Traction Control: YES
- Multiple Speed Gear levels: Low, Medium, High, Reverse
- Integrated Smart Switch Function: supports sliding / rolling on and automatic shutdown function; supports modification of automatic shutdown time (standby current as low as 50uA)
- Two Power ON/OFF Switch Options: Self-resetting switch with green LED, button diameter: 16mm; Self-locking switch with red LED, button diameter: 16mm (When using the self-locking switch, please remember to OFF the "Rolling Start ON" switch. Otherwise, the ESC can not be powered off)
- Built-in IMU chip LSM6DS3TR-C (accelerometer, gyroscope)
- Power Wire: 8.0mm bullets female X 2pcs
- Motor Wire: 8.0mm bullets female X 3pcs
- Size: L140*W69.6*H32mm (including Alu. Case)

Features:

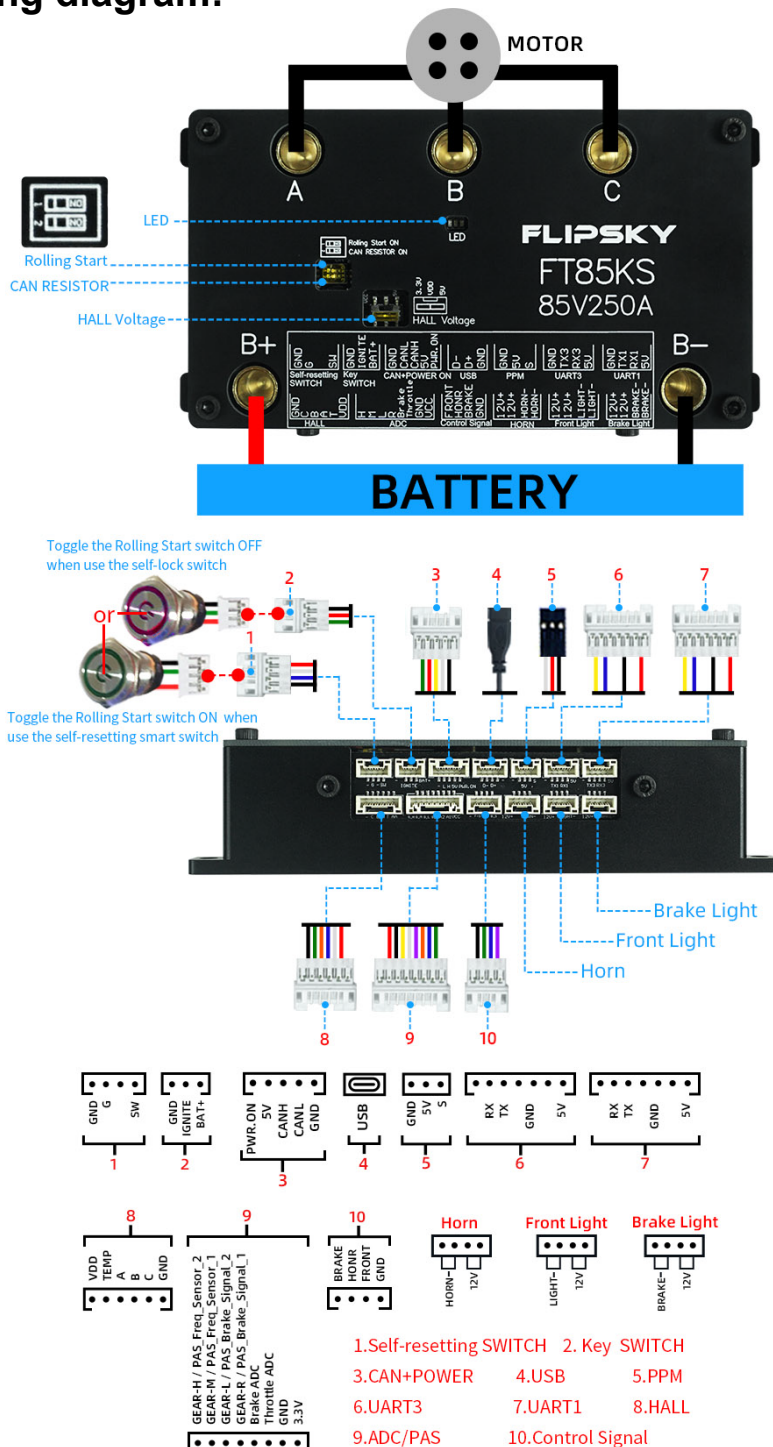
- Adopting ARM Cortex-M4 structure with the max 240MHz , improves PWM switching frequency and the motor RPM.
- Programmable via Flipsky ESC Tool to be compatible with multi-types of motor, different battery voltages and many control ways
- Programmable via Flipsky remote controller VX4Pro (remote firmware version: VX4Pro_V1.1 or above).
- Supporting motor FOC with low working noise and smooth starting.
- Supporting Real-time data and fault alerts.
- Supporting adaptive variable PWM frequency control

<https://flipsky.net/>

E-mail:flipskytech@outlook.com

- Supporting sensor mode(HALL, ENCODER) and sensorless mode(IPDS, HFPI,OPEN).
- Supporting control types: Current, Duty Cycle, Speed control, Position Control
- Protection functions: low and high voltage protection, over-current protection, temperature abnormally protection, MOSFET over-temperature and over current protection, battery over discharge protecting, motor over-temperature protection.
- Supporting extra interfaces like front lights, brake lights and horn fixtures (12V 3A) which can be controlled by remote VX4 or external switches.
- Supporting various motor temperature sensor types such as NTC-10K-25℃, NTC-10K-100℃, PT1000, PTC-1K-100℃, KTY and so on.

Wiring diagram:



To download the Flipsky ESC Tool and User Manual, please scan the QR code:



<https://flipsky.net/>

E-mail:flipskytech@outlook.com

FT85KS规格书（版本V1.5）

规格参数:

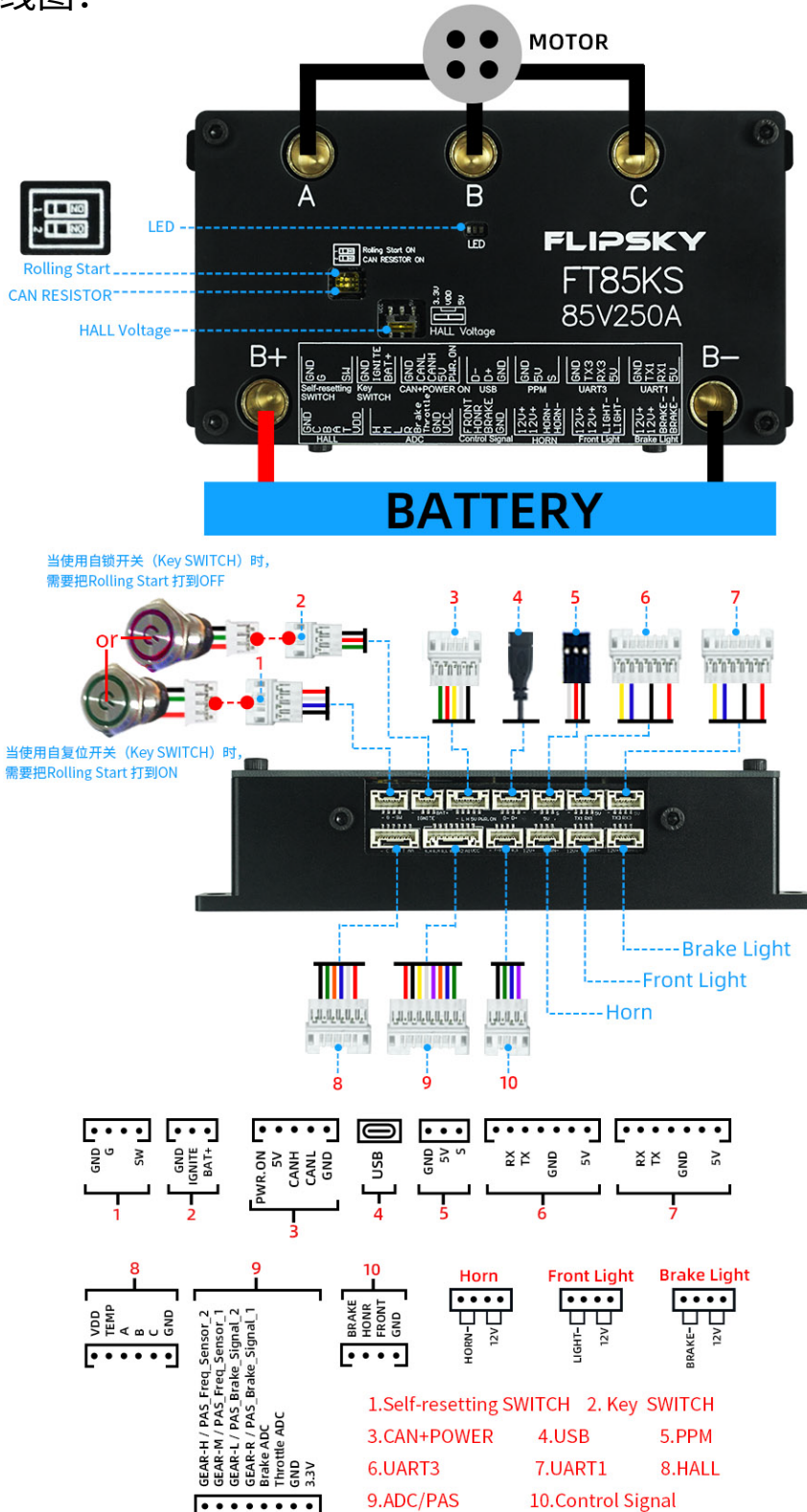
- 固件：V1.5及以上，支持通过Flipsky ESC Tool升级
- 电压：14V-85V (安全范围：4S-20S)
- 持续电流：250A（取决于电调安装位置以及设备环境的温度、湿度）
- 峰值电流：400A
- BEC:5V@1A
- 电力线缆：2个8.0镀金母头
- 电机线缆：3个8.0镀金母头
- 调参工具：Flipsky ESC TOOL
- PWM变频控制：YES
- 多挡位限速控制：YES(UART、ADC)
- 再生制动：YES
- 通讯端口：USB, CAN, UART, PPM, ADC
- 控制信号类型：PPM, ADC, PAS, UART（目前兼容 VX1、VX1Pro、VX2、VX3、VX3Pro、VX4Pro 等遥控器）
- 启动模式：IPDS(无感轻载), HFPI(无感重载), HALL, ENCODER (ABI编码器, AS5047绝对式编码器SPI接口), OPEN(无感开环启动)
- 牵引力控制：YES
- 速度档位：低速档, 中速档, 高速档, 倒车档
- 前照灯、刹车灯、喇叭装置：支持外接开关进行独立控制
- 集成智能开关功能：支持滑动开机、支持自动关机功能、支持修改自动关机时间（待机电流低至50uA）
- 开关可选择：自复位开关带绿色LED按钮直径：16mm或 自锁开关带红色LED按钮直径：16mm（使用自锁开关按钮时，需要将“Rolling Start ON”设置到OFF，否则关不了机）
- 控制板内置 IMU 芯片- LSM6DS3TR-C (加速度计、陀螺仪)(目前还在研发阶段，待升级固件后才可以正常使用)
- 尺寸：长140mm、宽69.6mm、高32mm（带外壳）

产品特点:

- 采用ARM Cortex-M4架构，高达240MHz主频，可提高PWM控制频率和电机转速上限
- 可通过Flipsky ESC Tool编程，兼容多种电机、不同电池电压和多种控制方式
- 支持 Flipsky VX4Pro 遥控器调参（固件版本：VX4Pro_V1.1 以上版本）；
- 支持 FOC，工作噪音低，启动平稳；
- 支持实时数据和故障报警支持自适应PWM变频控制
- 支持传感器模式（HALL、ENCODER）和无传感器模式（IPDS、HFPI、OPEN）

- 支持控制类型：电流（Current），占空比（Duty Cycle），速度控制（Speed Control），位置控制（Position Control）
- 保护功能：低压保护；高压保护；电机过温保护；MOS管过流保护；电池过放保护；MOS管过温保护；集成3组12V功率电源，支持前照灯、刹车灯、喇叭装置（12V电源最大输出电流3A）；
- 支持多种类型电机温度传感器类型，包括NTC-10K-25°C、NTC-10K-100°C、PT1000、PTC-1K-100°C、KTY系列等多种类型温度传感器

接线图：



Flipsky ESC TOOL
软件+中文用户手册
下载二维码：

